Harshita Poojary

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Education

University of Southern California, Los Angeles, CA. **Aug 2023 - May 2025**
Master of Science - Computer Science **GPA: 3.81/4.00**
Coursework: Analysis of Algorithms, Deep Learning, Artificial Intelligence, Machine Learning, Natural Language Processing

University of Mumbai, Mumbai, India. **Jul 2015 - May 2019**
Bachelor of Engineering - Computer Engineering **GPA: 3.83/4.00**
Coursework: Data Structures, Soft Computing, Distributed Databases, Web Technologies

Technical Skills

Core Programming Languages: Python
Other Programming Languages: JavaScript, TypeScript, Go, C++, Java, HTML/CSS, Scala
Frameworks & Libraries: Node.js, AngularJS, PySpark, PyTorch, Keras, TensorFlow, OpenCV, Streamlit
Databases: MongoDB, Redis, MySQL, ElasticSearch, Milvus, Hadoop, DynamoDB
Tools & Platforms: Docker, Kubernetes, AWS, Azure, Jenkins, Git, Jira, Postman, Swagger, PowerBI, Tableau, Grafana, Prometheus, Kibana,

Work Experience

AI Intern **Jul 2025 – Present**
Deep Defense Solutions (Remote) *USA*
- Developing an LLM-orchestrated **hoax-probability** system combining OCR with hybrid ML modules, improving structured extraction using supervised fine-tuning (SFT), instruction tuning, RLHF, and CoT-based reasoning for more reliable incident classification.

Data Science Intern **Jun 2024 – Aug 2024**
Zero-True (Remote) *New York, USA*
- Delivered **10 interactive demos** of notebook-based ML applications by designing **SQL/Python-integrated dashboards** for statistical analysis and modeling, enabling stakeholders to dynamically analyze datasets.
- Enhanced usability by implementing **5** frontend components using **Vuetify** (dropdowns, checklists, etc.), expanding notebook functionality and improving client experience.
- Collaborated with **founders and stakeholders** to evaluate customer and developer feedback, prioritizing high-impact features that shaped the product roadmap and improved developer adoption.

Machine Learning Engineer **Dec 2021 – Jul 2023**
Reliance Jio *Mumbai, India*
- Collaborated with **NVIDIA researchers** to reduce face registration latency by **40%** by deploying RetinaFace models with **TensorRT FP16** model compression using **Python** and **C++** on **T4/A100 GPUs**, achieving real-time inference across 2 distributed clusters.
- Increased deployment velocity by **50%** by automating **CI/CD pipelines** with **Python, Node.js, Docker, Kubernetes, GitLab, and Azure**, embedding health and alerting across three ML systems.
- Achieved **90%** spoof-detection accuracy and scaled deployments to **20+ sites** by fine-tuning **ResNet-32** on **H2O.ai**, integrating FastAPI endpoints with real-time monitoring and profiling for performance evaluation and drift detection.
- Supervised **4** interns in data collection, model validation, additional model iteration, and code reviews, ensuring reproducibility, well-documented workflows, adherence to coding standards, and production readiness.

Software Engineer **Jun 2019 – Dec 2021**
Reliance Jio *Mumbai, India*
- Boosted engagement by **20%** for **100M+** users by engineering scalable microservices and REST APIs (**Node.js**) integrated with **MongoDB, MySQL, Redis, and Elasticsearch.**
- Built large-scale article/video ingestion in **Node.js, Spark** and **SQL** deployed via **Kubernetes** on **Linux systems**, boosting throughput and aggregation efficiency by **35%** across multi-device delivery.
- Developed observability with **Grafana**, **Prometheus**, and **Kibana**, reducing API downtime by **15%** through proactive latency/error monitoring and scheduled alerts.
- Increased user engagement by **30%** by customizing content as per user preferences, content history, and locality, delivering local news, user-focused recommendations, and genre-based content.

Software Engineer **Jan 2019 – Jun 2019**
TaksyKraft (Remote) *Hyderabad, India*
- Developed real-time engagement prediction APIs in **Python** and **SQL**, improving targeting accuracy and increasing client ROI by **10%**.
- Shortened reporting cycles by **40%** by building **ETL pipelines** and **Tableau dashboards**, accelerating enterprise decision-making.

Projects

Multi-Agent Reinforcement Learning with SchedNet (PyTorch, MPE, LBF)
- Developed cooperative multi-agent reinforcement learning algorithms using **SchedNet** on **MPE** and **LBF** environments, leveraging differentiable communication, agent scheduling, and **CTDE** for coordinated policy learning.

Unified Semantic Space for Multimodal Retrieval (PyTorch, Python, LanceDB)
- Designed a multimodal retrieval framework aligning **BridgeTower** text–image embeddings for video-based Q&A, integrating **LanceDB** for sub-second vector search with ranking and deploying an interactive **Streamlit** interface.

Comparison of Model Performance with Knowledge Distillation and Quantization (PyTorch, Python)
- Fine-tuned **LLaMA-3-8B** with **4-bit** quantization and LoRA-based logit distillation, incorporating **Chain-of-Thought (CoT)** prompting to improve reasoning consistency and benchmarking performance on GSM8K (8-shot), achieving **76.9%** accuracy with low inference cost.